

The Three-Generation Map

Complete Order-One Classification of the Finite Spectral Triple

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2026-09-27 — v1.0

Research report. Numerical results (T4) unless marked machine-proven (T3). Not peer-reviewed.

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Abstract

We report the complete classification of the order-one nullspace of the finite spectral triple Dirac operator for three generations ($H_F = C^{96} = C^{32} \times C^3$). The order-one condition $[[D_F, \pi(a)], \pi \deg(b)] = 0$ admits a 402-real-dimensional space of Dirac operators: 3×46 generation-diagonal plus 3×88 generation-mixing directions (residuals $\sim 10^{-15}$). Every direction is classified by support, species, and generation structure. Applying charge conservation and one-Higgs minimality selects 84 directions: 4×18 Yukawa plus 12 Majorana — exactly the Standard Model flavour parameter count. The triplication is imposed, not derived; flavour structure is input. Order-one admits; physics selects; the count matches.

1. The framework

The finite spectral triple $(A_F, H_F, D_F, J, \Gamma)$ with $A_F = C \oplus H \oplus M_3(C)$ encodes the Standard Model's internal geometry. Two consistency conditions govern the Dirac operator:

- Order-zero: $[\pi(a), \pi \deg(b)] = 0$ — the left and right actions commute. -
- Order-one: $[[D_F, \pi(a)], \pi \deg(b)] = 0$ — the Dirac operator is a first-order differential operator in the noncommutative sense.

Machine-proven in Lean 4 (T3): order-zero for all 144 selected generator pairs, the KO signs ($J^2 = 1, J \Gamma = -\Gamma J$), full unitality, and order-one for the imposed one-generation Dirac ansatz (all 144 pairs, zero sorrys). The Dirac ansatz

itself — the five-block Yukawa/Majorana structure — is imposed as physical input, not derived.

2. One generation: 46 directions

Scanning the full 24-real-dimensional generator algebra (T4 numerical), the admissible Dirac space is 272 real-dimensional, and the order-one nullspace is 46-dimensional. Classified (DIRECTIONS_36.md):

Family	Real dims	Fate
SM Yukawa + Majorana	10	selected (physical input)
Flipped Yukawas	8	shelved (one-Higgs minimality)
Exotic Majorana	28	27 killed by charge conservation

Structural invariants: C-block numerically zero; $B = \text{conj}(A)$; the 36 extras are grading-odd, J-compatible, self-adjoint — genuine order-one directions, not artifacts. Verdict: order-one does not select the SM; it admits a 46-dimensional space in which the SM's 10 directions sit.

3. Triplication: C^{96} and the 402 census

Imposing three generations, $H_F = C^{32} (x) C^3$, with arbitrary 3x3 complex Yukawa matrices (Y_{ν}, Y_e, Y_u, Y_d) and a symmetric 3x3 Majorana matrix:

- Admissible Dirac space: 2352 real dimensions ($= 3 \times 272 + 3 \times 512$). - Order-one nullity: $402 = 3 \times 46$ (diagonal) + 3×88 (mixing). - Verified over 3 random trials: grading-oddness, self-adjointness, J-compatibility, and order-one all return exact zeros (residuals $\sim 10^{-15}$); the tests are non-vacuous (probed commutators of order ~ 17 -19).

The count 402 — not $9 \times 46 = 414$ — confirms the mixing structure is nontrivial: generation-mixing is constrained differently than naive triplication would suggest.

4. Classifying the 264 mixing directions

Per unordered generation pair, the 88 real mixing dimensions form a direct sum of slot-disjoint subspaces (joint projection residual 4.7×10^{-15}):

Family	Real dims	Complex	Fate
SM-like mixing	18	8 Yukawa + 1 MR	survives

Family	Real dims	Complex	Fate
Flipped Yukawa mixing	16	8	shelved (one-Higgs minimality)
Exotic Majorana mixing	54	27	killed by charge conservation

The 18 SM-like mixing directions per pair are the off-diagonal entries of the flavour matrices — the CKM/PMNS-type directions. Structural facts: $B = \text{conj}(A)$ exactly; $C = 0$ forced; colour locking holds in all sectors; E-symmetry drops in the mixing sector ($\max ||E - E^T|| = 1.15$, vs exactly 0 in one generation) with an "L-shaped" support of 28 entries. Direct order-one recheck: max residual 1.62×10^{-15} over 88×576 pairs.

No genuinely new structures appear: the mixing classification is the one-generation verdict crossed with 3×3 flavour. Yukawa parameters double via $(g, g')/(g', g)$ ordering ($16 \rightarrow 32$); the Majorana sector relaxes ($30 \rightarrow 56$).

5. The selection: 84

Sector	Directions	Count
Diagonal SM (3×10)	Yukawa + Majorana per generation	30
Mixing SM-like (3×18)	off-diagonal flavour entries	54
Total selected		84

$84 = 4 \times 18$ (four 3×3 complex Yukawa matrices) + 12 (symmetric 3×3 M_R) — exactly the Standard Model flavour parameter count. Order-one admits 402; charge conservation plus one-Higgs minimality selects 84; the Standard Model uses 84.

6. Numerical rigor

- All residuals at or below $\sim 10^{-14}$; direct rechecks at $\sim 10^{-15}$. - Two measurement artifacts were caught and corrected by an independent verification script before publication (sector-rank overlap bug; complex pattern construction bug). Corrected values are recorded with notes; uncorrected numbers are not trusted. - Brute-force 96×96 enumeration was used where structural reduction was unavailable; the user authorized brute force explicitly.

7. Honest ledger

Claim	Tier	Status
Order-zero, 144/144 pairs	T3	machine-proven (Lean 4, zero sorrys)
Order-one, one-gen ansatz	T3	machine-proven (Lean 4, zero sorrys)
46-dim nullspace, one gen	T4	numerical (full 24-generator scan)
402-dim nullspace, three gen	T4	numerical (3 random trials)
264-direction classification	T4	numerical (residuals $\sim 10^{-15}$)
Triplication to 3 generations	imposed	not derived from the axioms
Flavour structure (CKM/PMNS)	input	not derived; arbitrary matrices
"Order-one selects the SM"	false	order-one admits; physics selects

8. Next steps

1. Composition theorem for thet-language v2 (order-one analog for composed thet events), precisely stated — K3. 2. Second instantiation of the repeat operator R beyond the finite triple — K1. 3. Spectral action / EFT over the selected 84-direction space. 4. Pre-registered harmonic series for one mass ratio, or permanent quarantine of the mass-ratio program — K2.

Repository: github.com/jeffgurd888/thet-logos. Reports: [attack4_corrected/DIRECTIONS_36.md](#), [THREE_GEN.md](#), [MIXING_264.md](#). Lean sources: [lean/ThetLogos/MartinettiRep.lean](#), [ThreeGen.lean](#).